

standard deviations, a random variable will be expected to be just 10.

In Extremistan, things work differently. The conditional expectation of an increase in a random variable does not converge to the threshold as the variable gets larger. In the real world, say with stock returns (and all economic variables), conditional on a loss being worse than 5 units, using any unit of measure (it makes little difference), it will be around 8 units. Conditional that a move is more than 50 units, it should be around 80 units, and if we go all the way until the sample is depleted, the average move worse than 100 units is 250 units! This extends to all areas in which I found sufficient samples. This tells us that there is “no” typical failure and “no” typical success. You may be able to predict the occurrence of a war, but you will not be able to gauge its effect! Conditional on a war killing more than 5 million people, it should kill around 10 million (or more). Conditional on it killing more than 500 million, it would kill a billion (or more, we don’t know). You may correctly predict that a skilled person will get “rich,” but, conditional on his making it, his wealth can reach \$1 million, \$10 million, \$1 billion, \$10 billion—there is no typical number. We have data, for instance, for predictions of drug sales, conditional on getting things right. Sales estimates are totally uncorrelated to actual sales—some drugs that were correctly predicted to be successful had their sales underestimated by up to 22 times.

This absence of “typical” events in Extremistan is what makes something called prediction markets (in which people are assumed to make bets on events) ludicrous, as they consider events to be binary. “A war” is meaningless: you need to estimate its damage—and no damage is typical. Many predicted that the First World War would occur, but nobody really predicted its magnitude. One of the reasons economics does not work is that the literature is almost completely blind to this point.

Accordingly, Ferguson’s methodology (mentioned in [Chapter 1](#)) in looking at the prediction of events as expressed in the price of war bonds is sounder than simply counting predictions, because a bond, reflecting the costs to the governments involved in a war, is priced to cover the probability of an event *times* its consequences, not just the probability of an event. So we should not focus on whether someone “predicted” an event without his statement having consequences attached to it.

Associated with the previous fallacy is the mistake of thinking that my message is that these Black Swans are necessarily more probable than assumed by conventional methods. They are mostly *less* probable, but have bigger effects. Consider that, in a winner-take-all environment,